

A proof of the conjectured linear recurrence and generating function for OEIS A281209

Adrian Perez Fontelles and Gaspard Moulinier, Independent researchers

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1 The conjecture

The Online Encyclopedia of Integer Sequences records [A281209](#) as

Number of 5 X n 0..1 arrays with no element equal to more than one of its horizontal and antidiagonal neighbors, with the exception of exactly one element, and with new values introduced in order 0 sequentially upwards

and carries in its Formula section the lines:

Empirical: $a(n) = 4*a(n-1) - 4*a(n-2) - 2*a(n-3) + 4*a(n-4) - a(n-6)$ for $n > 11$.

Empirical g.f.: $x^2*(130 - 72*x - 596*x^2 + 489*x^3 + 422*x^4 - 363*x^5 - 56*x^6 + 114*x^7 + 10*x^8 - 4*x^9) / ((1 - x)^2*(1 - x - x^2)^2)$. -
Colin Barker, Feb 18 2019

That line carries no separate signature; the entry itself is by R. H. Hardin (Jan 17 2017).

The line quoted ending in that signature is contributed by Colin Barker on Feb 18 2019.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 7, last modified Feb 18 2019, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 4a(n-1) - 4a(n-2) - 2a(n-3) + 4a(n-4) - a(n-6).$$

stated for $n > 11$.

Written out, the claim is

$$\sum_{n \geq 1} a(n) x^n = \frac{130x^2 - 72x^3 - 596x^4 + 489x^5 + 422x^6 - 363x^7 - 56x^8 + 114x^9 + 10x^{10} - 4x^{11}}{1 - 4x + 4x^2 + 2x^3 - 4x^4 + x^6}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 5$ array over $\{0, \dots, 1\}$ with $n' = n$ rows, the entry's array transposed so the growing direction is downwards, the neighbour offsets transposed with it.

The entry's neighbours, *horizontal and antidiagonal*, are the cells at the offsets

$$\mathcal{N} = \{(-1, 0), (-1, 1), (1, -1), (1, 0)\}$$

that lie inside the array, and $c(i, j)$ is the number of them carrying the same value as (i, j) .

Call a cell *bad* when $c(i, j) > 1$.

The condition is that exactly 1 cells are bad.

The entry's further clause says that reading the array row by row the values $0, \dots, 1$ first occur in increasing order.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n columns.

3 The count is a walk count

Every neighbour offset moves at most one row, so whether a cell of row i is bad is decided by rows $i - 1$, i and $i + 1$. Take as state the last two rows, the number of bad cells so far capped at 2, and a counter for how many values have been introduced. Appending a row decides the row before it, and the accepting condition decides the last one.

An array of n' rows is then exactly a walk of length n' from the empty state to an accepting one, so $a(n) = w^T M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 1800 states, 476 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 41$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of that 41×41 matrix; the bound is computed, not assumed.

5 The decision procedure

Let $c_1, \dots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^D c_i a(n - i)$, let $q(t) = t^D - \sum_{i=1}^D c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m = \tilde{w}^T L^{m-1} q(L) \tilde{f} \quad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a , one gets $u_m = a(n) - \sum_{i=1}^D c_i a(n - i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^\top L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L , of degree 41: writing that polynomial as $t^{41} - \sum_{i=1}^{41} \lambda_i t^{41-i}$ gives $u_{m+41} = \sum_{i=1}^{41} \lambda_i u_{m+41-i}$. Hence 41 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A281209, the recurrence $a(n) = 4a(n-1) - 4a(n-2) - 2a(n-3) + 4a(n-4) - a(n-6)$ holds for every $n > 11$, and fails at $n = 11$.*

Proof. The residuals u_m were computed exactly for $m = 1, \dots, 83$. The last nonvanishing one is at $m = 5$, that is at $n = 11$. After it, more than $S = 41$ consecutive residuals vanish, so by Section 5 every later residual vanishes as well. $\square$

The entry claims the recurrence for $n > 11$. The theorem is at least as strong.

Theorem 2. *For A281209, the generating function is exactly $\sum_{n \geq 1} a(n)x^n = \frac{130x^2 - 72x^3 - 596x^4 + 489x^5 + 422x^6 - 363x^7 - 130x^8 + 72x^9 + 596x^{10} - 489x^{11} - 422x^{12} + 363x^{13} - 130x^{14}}{1 - 4x + 4x^2 + 2x^3 - 4x^4 + 4x^5 - 4x^6 + 2x^7 - 4x^8 + 4x^9 - 4x^{10} + 2x^{11} - 4x^{12} + 4x^{13} - 4x^{14} + 2x^{15} - 4x^{16} + 4x^{17} - 4x^{18} + 2x^{19} - 4x^{20} + 4x^{21} - 4x^{22} + 2x^{23} - 4x^{24} + 4x^{25} - 4x^{26} + 2x^{27} - 4x^{28} + 4x^{29} - 4x^{30} + 2x^{31} - 4x^{32} + 4x^{33} - 4x^{34} + 2x^{35} - 4x^{36} + 4x^{37} - 4x^{38} + 2x^{39} - 4x^{40} + 4x^{41} - 4x^{42} + 2x^{43} - 4x^{44} + 4x^{45} - 4x^{46} + 2x^{47} - 4x^{48} + 4x^{49} - 4x^{50} + 2x^{51} - 4x^{52} + 4x^{53} - 4x^{54} + 2x^{55} - 4x^{56} + 4x^{57} - 4x^{58} + 2x^{59} - 4x^{60} + 4x^{61} - 4x^{62} + 2x^{63} - 4x^{64} + 4x^{65} - 4x^{66} + 2x^{67} - 4x^{68} + 4x^{69} - 4x^{70} + 2x^{71} - 4x^{72} + 4x^{73} - 4x^{74} + 2x^{75} - 4x^{76} + 4x^{77} - 4x^{78} + 2x^{79} - 4x^{80} + 4x^{81} - 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4x^{1004} + 4x^{1005} - 4x^{1006} + 2x^{1007} - 4x^{1008} + 4x^{1009} - 4x^{1010} + 2x^{1011} - 4x^{1012} + 4x^{1013} - 4x^{1014} + 2x^{1015} - 4x^{1016} + 4x^{1017} - 4x^{1018} + 2x^{1019} - 4x^{1020} + 4x^{1021} - 4x^{1022} + 2x^{1023} - 4x^{1024} + 4x^{1025} - 4x^{1026} + 2x^{1027} - 4x^{1028} + 4x^{1029} - 4x^{1030} + 2x^{1031} - 4x^{1032} + 4x^{1033} - 4x^{1034} + 2x^{1035} - 4x^{1036} + 4x^{1037} - 4x^{1038} + 2x^{1039} - 4x^{1040} + 4x^{1041} - 4x^{1042} + 2x^{1043} - 4x^{1044} + 4x^{1045} - 4x^{1046} + 2x^{1047} - 4x^{1048} + 4x^{1049} - 4x^{1050} + 2x^{1051} - 4x^{1052} + 4x^{1053} - 4x^{1054} + 2x^{1055} - 4x^{1056} + 4x^{1057} - 4x^{1058} + 2x^{1059} - 4x^{1060} + 4x^{1061} - 4x^{1062} + 2x^{1063} - 4x^{1064} + 4x^{1065} - 4x^{1066} + 2x^{1067} - 4x^{1068} + 4x^{1069} - 4x^{1070} + 2x^{1071} - 4x^{1072} + 4x^{1073} - 4x^{1074} + 2x^{1075} - 4x^{1076} + 4x^{1077} - 4x^{1078} + 2x^{1079} - 4x^{1080} + 4x^{1081} - 4x^{1082} + 2x^{1083} - 4x^{1084} + 4x^{1085} - 4x^{1086} + 2x^{1087} - 4x^{1088} + 4x^{1089} - 4x^{1090} + 2x^{1091} - 4x^{1092} + 4x^{1093} - 4x^{1094} + 2x^{1095} - 4x^{1096} + 4x^{1097} - 4x^{1098} + 2x^{1099} - 4x^{1100} + 4x^{1101} - 4x^{1102} + 2x^{1103} - 4x^{1104} + 4x^{1105} - 4x^{1106} + 2x^{1107} - 4x^{1108} + 4x^{1109} - 4x^{1110} + 2x^{1111} - 4x^{1112} + 4x^{1113} - 4x^{1114} + 2x^{1115} - 4x^{1116} + 4x^{1117} - 4x^{1118} + 2x^{1119} - 4x^{1120} + 4x^{1121} - 4x^{1122} + 2x^{1123} - 4x^{1124} + 4x^{1125} - 4x^{1126} + 2x^{1127} - 4x^{1128} + 4x^{1129} - 4x^{1130} + 2x^{1131} - 4x^{1132} + 4x^{1133} - 4x^{1134} + 2x^{1135} - 4x^{1136} + 4x^{1137} - 4x^{1138} + 2x^{1139} - 4x^{1140} + 4x^{1141} - 4x^{1142} + 2x^{1143} - 4x^{1144} + 4x^{1145} - 4x^{1146} + 2x^{1147} - 4x^{1148} + 4x^{1149} - 4x^{1150} + 2x^{1151} - 4x^{1152} + 4x^{1153} - 4x^{1154} + 2x^{1155} - 4x^{1156} + 4x^{1157} - 4x^{1158} + 2x^{1159} - 4x^{1160} + 4x^{1161} - 4x^{1162} + 2x^{1163} - 4x^{1164} + 4x^{1165} - 4x^{1166} + 2x^{1167} - 4x^{1168} + 4x^{1169} - 4x^{1170} + 2x^{1171} - 4x^{1172} + 4x^{1173} - 4x^{1174} + 2x^{1175} - 4x^{1176} + 4x^{1177} - 4x^{1178} + 2x^{1179} - 4x^{1180} + 4x^{1181} - 4x^{1182} + 2x^{1183} - 4x^{1184} + 4x^{1185} - 4x^{1186} + 2x^{1187} - 4x^{1188} + 4x^{1189} - 4x^{1190} + 2x^{1191} - 4x^{1192} + 4x^{1193} - 4x^{1194} + 2x^{1195} - 4x^{1196} + 4x^{1197} - 4x^{1198} + 2x^{1199} - 4x^{1200} + 4x^{1201} - 4x^{1202} + 2x^{1203} - 4x^{1204} + 4x^{1205} - 4x^{1206} + 2x^{1207} - 4x^{1208} + 4x^{1209} - 4x^{1210} + 2x^{1211} - 4x^{1212} + 4x^{1213} - 4x^{1214} + 2x^{1215} - 4x^{1216} + 4x^{1217} - 4x^{1218} + 2x^{1219} - 4x^{1220} + 4x^{1221} - 4x^{1222} + 2x^{1223} - 4x^{1224} + 4x^{1225} - 4x^{1226} + 2x^{1227} - 4x^{1228} + 4x^{1229} - 4x^{1230} + 2x^{1231} - 4x^{1232} + 4x^{1233} - 4x^{1234} + 2x^{1235} - 4x^{1236} + 4x^{1237} - 4x^{1238} + 2x^{1239} - 4x^{1240} + 4x^{1241} - 4x^{1242} + 2x^{1243} - 4x^{1244} + 4x^{1245} - 4x^{1246} + 2x^{1247} - 4x^{1248} + 4x^{1249} - 4x^{1250} + 2x^{1251} - 4x^{1252} + 4x^{1253} - 4x^{1254} + 2x^{1255} - 4x^{1256} + 4x^{1257} - 4x^{1258} + 2x^{1259} - 4x^{1260} + 4x^{1261} - 4x^{1262} + 2x^{1263} - 4x^{12$*

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